

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

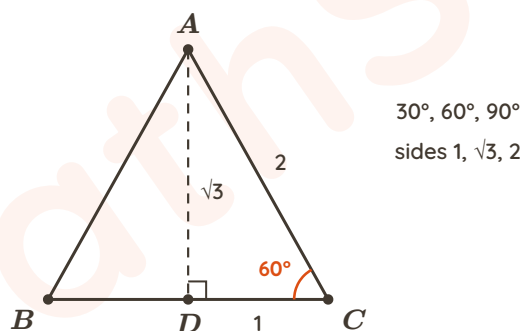
Chapter 23: Trigonometrical Ratios of Standard Angles

Topic-wise short notes with solved examples

The values of the trigonometrical ratios of 0° , 30° , 45° , 60° and 90° , how they are obtained, and their use in evaluating expressions and solving simple trigonometric equations.

1 Where the values come from

- 45° : in an isosceles right triangle with equal sides 1, the hypotenuse is $\sqrt{2}$. So $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$ and $\tan 45^\circ = 1$.
- 30° and 60° : in an equilateral triangle of side 2, an altitude makes two right triangles with angles 30° , 60° , 90° and sides 1, $\sqrt{3}$, 2.
- So $\sin 30^\circ = \frac{1}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$.



Half of an equilateral triangle of side 2

2 Table of values

Angle	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined
cosec	not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	not defined
cot	not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

- **Memory aid for sine:** $\sqrt{\frac{0}{4}}, \sqrt{\frac{1}{4}}, \sqrt{\frac{2}{4}}, \sqrt{\frac{3}{4}}, \sqrt{\frac{4}{4}}$. The cosine row is the sine row reversed.
- As the angle increases from 0° to 90° : **sin** and **tan increase**; **cos decreases**.
- $\sin \theta = \cos \theta$ only at $\theta = 45^\circ$.

3 Evaluating expressions

- Replace each ratio by its value, then simplify. Remember $\sin^2 30^\circ$ means $(\sin 30^\circ)^2$.

Example 3.1 Evaluate $\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ$.

SOLUTION

$$\frac{1}{2} \times \frac{1}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} = \frac{1}{4} + \frac{3}{4}$$

1

Example 3.2 Evaluate $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$.

SOLUTION

$$2 + \frac{3}{4} - \frac{3}{4}$$

2

Example 3.3 Evaluate $\frac{\tan 60^\circ - \tan 30^\circ}{1 + \tan 60^\circ \tan 30^\circ}$.

SOLUTION

$$\frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{1 + 1} = \frac{\frac{2}{\sqrt{3}}}{2}$$

$$\frac{1}{\sqrt{3}}, \text{ which is } \tan 30^\circ$$

Example 3.4 Evaluate $4 \sin^2 60^\circ + 3 \tan^2 30^\circ - 8 \sin 45^\circ \cos 45^\circ$.

SOLUTION

$$4 \times \frac{3}{4} + 3 \times \frac{1}{3} - 8 \times \frac{1}{2}$$

0

4 Verifying results

- Substitute the given angle on both sides and check that the values agree.
- **Warning:** $\sin(A + B)$ is **not** $\sin A + \sin B$, and $\sin 2A$ is **not** $2 \sin A$.

Example 4.1 If $A = 30^\circ$, verify that $\sin 2A = 2 \sin A \cos A$.

SOLUTION

$$\text{LHS} = \sin 60^\circ = \frac{\sqrt{3}}{2}. \text{ RHS} = 2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2}.$$

Both equal $\frac{\sqrt{3}}{2}$

Example 4.2 If $A = 60^\circ$ and $B = 30^\circ$, verify that $\cos(A - B) = \cos A \cos B + \sin A \sin B$.

SOLUTION

$$\text{LHS} = \cos 30^\circ = \frac{\sqrt{3}}{2}. \text{ RHS} = \frac{1}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \times \frac{1}{2}.$$

Both equal $\frac{\sqrt{3}}{2}$

Example 4.3 Show that $\sin 60^\circ \neq 2 \sin 30^\circ$.

SOLUTION

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \text{ but } 2 \sin 30^\circ = 1$$

They are different

5 Solving for an angle

- Match the given value with the table. In this chapter the angles lie between 0° and 90° .
- If the angle is 2θ or $\theta + 15^\circ$ and so on, find that angle first and then solve for θ .

Example 5.1 Find the acute angle θ if $2 \sin \theta = \sqrt{3}$.

SOLUTION

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\theta = 60^\circ$$

Example 5.2 Find θ if $\tan 3\theta = 1$ and 3θ is acute.

SOLUTION

$$3\theta = 45^\circ$$

$$\theta = 15^\circ$$

Example 5.3 Solve $2 \cos^2 \theta - 1 = 0$ for an acute angle.

SOLUTION

$$\cos^2 \theta = \frac{1}{2}, \text{ so } \cos \theta = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

Example 5.4 Find A and B if $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, where $A + B \leq 90^\circ$ and $A > B$.

SOLUTION

$$A + B = 90^\circ \text{ and } A - B = 30^\circ$$

$$A = 60^\circ, B = 30^\circ$$

Example 5.5 Solve $2 \sin^2 \theta - 3 \sin \theta + 1 = 0$ for $0^\circ < \theta \leq 90^\circ$.

SOLUTION

$$(2 \sin \theta - 1)(\sin \theta - 1) = 0, \text{ so } \sin \theta = \frac{1}{2} \text{ or } \sin \theta = 1$$

$$\theta = 30^\circ \text{ or } \theta = 90^\circ$$

6 Finding sides

Example 6.1 In triangle ABC, $\angle B = 90^\circ$, $\angle A = 30^\circ$ and $AC = 12$ cm. Find BC and AB.

SOLUTION

$$BC = AC \sin 30^\circ = 6 \text{ cm and } AB = AC \cos 30^\circ = 6\sqrt{3} \text{ cm}$$

$$6 \text{ cm and } 6\sqrt{3} \text{ cm}$$

7 Common mistakes

- Swapping the values of $\sin 30^\circ$ and $\sin 60^\circ$.
- Writing $\tan 90^\circ = 0$ or $\cot 0^\circ = 0$. They are not defined.
- Treating $\sin(A + B)$ as $\sin A + \sin B$.
- Reading $\sin^2 45^\circ$ as $\sin 90^\circ$.
- Giving θ when the equation was solved for 2θ or 3θ .
- Leaving a surd in the denominator when a rationalised answer is expected.